Project: Local Multimodal Hybrid RAG Agent
Class: CSCI E-89, Deep Learning, Harvard Extension School, Fall ‘25
Professors: Prof. Zoran B. Djordjević & Prof. Rahul Joglekar
Student: Robert Frenette

Abstract

Traditional chatbots face significant limitations when handling complex, domain-specific queries that require access to both structured data (DBs) and unstructured documents (DOCs, PDFs). Standard LLMs suffer from hallucinations, lack access to real-time data, and cannot seamlessly integrate multiple knowledge sources. Also, cloud-based AI services pose critical security and compliance risks for industries handling sensitive information (financial services, healthcare, insurance) where customer data, proprietary policies, and confidential documents cannot be transmitted over the public internet due to regulatory requirements (HIPAA, PCI) and data governance concerns.

This project demonstrates a hybrid retrieval augmented generation (RAG) Agent that intelligently routes user queries between a SQL database (SQLite) and unstructured document retrieval (ChromaDB vector store), providing source-grounded responses while maintaining complete data privacy through local (Ollama) processing.

Dataset

This project utilizes two complementary datasets for a fictitious insurance company (AI Agent Insure). The first is a structured SQLite DB containing 100 insureds with 208 policies, 46 claims, and associated coverage details, risk profiles, and claims history in five related tables. The other is an unstructured knowledge base of 20+ PDF documents organized into categories such as company information, agent support guides, procedural documentation, and personnel information. These PDFs are processed into vector embeddings using nomic-embed-text, then put into a ChromaDB vector store with enhanced metadata (document type, chunk indices, and page counts).

Technology and Features Demonstrated

The application showcases a hybrid RAG architecture built with LangChain, running entirely on local infrastructure using Ollama for LLM inference (llama3.2:3b) and embeddings (nomic-embed-text).

Key features include: Intelligent query routing that automatically detects whether queries require SQL database operations or RAG document retrieval; a complete RAG pipeline with optimized retrieval (top-k=4), custom prompt engineering to minimize hallucinations, and summary-based conversational memory for multi-turn dialogues; multimodal interaction supporting both text and voice input (OpenAI Whisper) with audio output (text-to-speech); and enhanced source attribution with document metadata, enabling users to verify information origins.

Resources

2 min (short) video: https://youtu.be/HinayJJ687k
15 min (long) video: https://youtu.be/V0mAQjiUK6Q
YouTube Project Playlist: https://www.youtube.com/playlist?list=PLRBkbp6t5gM12JGRlbGomEETO2RknoRbP
GitHub Repo: https://github.com/RobertFrenette/CSCI_E-89_Deep_Learning_Fall_25’